

Q.1) Generate parent process to write unnamed pipe and will read from it

```
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <stdlib.h>

int main() {
    int fd[2]; // fd[0] = read end, fd[1] = write end
    char write_msg[] = "Hello from parent process!";
    char read_msg[100];

    // Create unnamed pipe
    if (pipe(fd) == -1) {
        perror("pipe");
        exit(1);
    }

    // Parent writes to pipe
    write(fd[1], write_msg, strlen(write_msg) + 1); // +1 for null terminator
    printf("Parent wrote to pipe: %s\n", write_msg);

    // Close write end before reading
    close(fd[1]);

    // Parent reads from pipe
    read(fd[0], read_msg, sizeof(read_msg));
    printf("Parent read from pipe: %s\n", read_msg);

    // Close read end
    close(fd[0]);

    return 0;
}
```

Q.2) Write a C program to Identify the type (Directory, character device, Block device, Regular file, FIFO or pipe, symbolic link or socket) of given file using stat() system call.

```
#include <stdio.h>

#include <stdlib.h>

#include <sys/stat.h>

#include <unistd.h>
```

```
void printFileType(mode_t mode) {
    if (S_ISREG(mode))
        printf("Regular file\n");
    else if (S_ISDIR(mode))
        printf("Directory\n");
    else if (S_ISCHR(mode))
        printf("Character device\n");
    else if (S_ISBLK(mode))
        printf("Block device\n");
    else if (S_ISFIFO(mode))
        printf("FIFO or pipe\n");
    else if (S_ISLNK(mode))
        printf("Symbolic link\n");
    else if (S_ISSOCK(mode))
        printf("Socket\n");
    else
        printf("Unknown file type\n");
}
```

```
int main(int argc, char *argv[]) {
    if (argc != 2) {
        fprintf(stderr, "Usage: %s <filename>\n", argv[0]);
        exit(1);
    }
```

```
    struct stat fileStat;
```

```
if (stat(argv[1], &fileStat) < 0) {  
    perror("stat");  
    exit(1);  
}  
  
printf("File: %s\n", argv[1]);  
printf("File type: ");  
printFileType(fileStat.st_mode);  
  
return 0;  
}
```